

Transcript identity changes fusion junction antisense comparisons in extraskeletal myxoid chondrosarcoma

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To the Editor,

Antisense oligonucleotides targeting fusion junctions require exact transcript sequences. In extraskeletal myxoid chondrosarcoma (EMC), we found that resolving published transcript annotations establishes correspondence with an existing design, while including additional normal-parent isoforms changes specific sequence comparisons. This audit adds an attributable model crosswalk and accession-specific findings to established antisense specificity principles.[1,2]

We examined primary model descriptions and seven deposited fusion records, which support five distinct junctions rather than seven independent patients. Thirty-five existing 16-nucleotide designs cover seven junctions: five supported by fusion deposits, one reconstructed from patient-model coordinates and one previously reconstructed from an engineered-construct description. Five binding positions per junction move the seam within the proposed six-base DNA gap, flanked by five locked nucleic acid residues on either side. For each design, we measured the longest consecutive exact match to a nonfusion RNA window spanning all six gap positions. The comparator expanded from six parent transcripts to 77 records representing 71 distinct sequences across the same six genes. Supplementary Methods provides the source classifications, algorithms and all positions.

Bangerter et al. label USZ20-EMC1 as EWSR1 exon 13 joined to NR4A3 exon 2.[1] Coordinates and protein identifiers in their Figure 4b resolve to GRCh37 transcripts ENST00000414183.2 and ENST00000330847.1. The corresponding boundaries are exon 12 of NM_005243.4 and exon 3 of NM_006981.4. Thirty-base flanks agree between transcript and genomic sequences and match the catalogue's exon-12/exon-3 junction. The centred reconstructed antisense sequence is GGGCATATCATCAAAC. Transferring the labels “13/2” literally to the catalogue's different transcripts gives AGTGGGCTCTCCACGG, differing at 11 of 16 positions, including four gap positions. This resolves reference correspondence and corrects our earlier interpretation that the published figure was insufficient. It does not establish the patient's RNA consensus. The companion USZ22-EMC2 junction remains unresolved because its displayed intervals and direction do not specify a unique seam.

Additional parent isoforms lengthen the gap-spanning match for 9 of 35 designs. For the centred TCF12-junction design GGGCATATCCATCAGA, the maximum increases from seven to eleven bases. Its target TCTGATGGATATGCCC and nonfusion window AAATGTGGATATGCCC share the final eleven bases, spanning the complete gap. This sequence occurs in NR4A3 NM_173200.3 and its source cDNA U12767.1, associated with the original MINOR report.[3] These records corroborate one sequence, not independent biological replications. Other changes involve FUS isoforms or a nonfusion NR4A3 splice. USZ20's five values remain 11, 7, 8, 8 and 8 bases as donor contribution increases from six to ten target bases. No complete 16-base match occurs in the expanded corpus.

These findings refine the interpretation of existing designs. The earlier 190-design catalogue mainly contains hypothetical junctions; this selected audit neither updates every design nor estimates patient coverage. Binding-position dependence and hybridization-related off-target concerns are already established.[2] The added information is which model annotation maps to which reference sequence, and which omitted normal isoforms change particular comparisons. Versioned sequences, evidence tables, all design comparisons and reproducible code are available in the accompanying archive.[4]

The limitations are shared across these results. USZ20's genome build and coordinate convention are inferred; patient variants or inserted bases remain unexcluded. The parent-gene corpus does not establish transcriptome-wide specificity or tissue expression. A partial, nonrandom sequencing-data probe supplied no additional junction confirmation; its negative result and limits remain in the supplement. No oligonucleotide was synthesized or tested. Exact-match lengths and the previously used ten-base cutoff are not validated biological thresholds. RNA cleavage, selectivity and safety require experimental evidence.

Declarations

The author directed this public-data study, which involved no new recruitment or experiments. No external funding or competing financial interests are declared; the author is an EMC survivor. Claude assisted the preceding resource; OpenAI Codex with GPT-6-Astra assisted retrieval, analysis, verification and drafting. Computational checks and an independent AI critique do not constitute experimental validation or human peer review. The author is responsible for the manuscript.

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